CONNECTIVITY MASTER UNIT [WITH CENTER DISPLAY]

id0920zz113400

Purpose, Function

- The connectivity master unit (CMU) controls the entire system such as communication between mobile devices and Bluetooth®, and the sending/receiving of video/audio signals from units related to the entertainment system.

Bluetooth® communication function

- The CMU uses the integrated Bluetooth® driver and antenna to perform communication between Bluetooth®-enabled devices and Bluetooth®.
- To perform Bluetooth® communication, pairing of a Bluetooth®-enabled device with the CMU is required to specify the device. For the pairing method, refer to the Bluetooth® PAIRING PROCEDURE in the workshop manual.

Specification

Item		Content
Compatible profiles	A2DP 1.0/1.2 (Advanced audio distribution profile)	A2DP is a profile for transmitting high quality audio data by means of a Bluetooth® device and wireless transmission technology.
	AVRCP 1.0/1.3/1.4 (Audio/video remote control profile)	AVRCP is a profile which provides a standard interface which controls television and audio devices. The function differs depending on the version.
Maximum number of programmable devices		7
Number of Bluetooth®-enabled devices which can be used simultaneously		2*1

*1 : A Hands-free call can be placed using another Bluetooth®-enabled device during audio playback using the other Bluetooth®-enabled device.

Note

- The profile indicates the wireless communication standard. If the profile of the Bluetooth®-enabled device differs from the CMU, wireless transmission may not be possible. In addition, depending on the profile version, the corresponding functions differ.

Operation signal reception function

- The CMU controls units related to the entertainment system based on the following operation signals which are sent in conjunction with the user operations from the operation signal output unit.

Operation signal output unit	Operation signal
Center display	LVDS signal (touch position operation signal)
Commander switch	Commander switch operation signal
Steering switch	Steering switch operation signal
Voice recognition microphone	Voice signal (voice command)

LVDS *2 signal conversion function

Note

- Communication between the CMU and the center display is performed via the LVDS communication line.

- The CMU converts signals sent from the following units to a LVDS signal and displays each type of information in the center display.

Output unit	Video signal/display request signal
Tuner and amplifier unit (TAU)	• TA (traffic announcement) information • Radio broadcast logo, album art, song name • Traffic information such as location and length of traffic jams
CD player	• Playback file information (artist/song name)
DVD/CD player	• DVD information (video) during playback
Auxiliary jack/USB port/SD card slot hub.	• Information for file being played (artist/song name)

Output unit	Video signal/display request signal
Devices such as Bluetooth®-enabled mobile phones and or mobile devices such as Smartphones/Bluetooth®-enabled devices	• Telephone book, incoming call record • Radio broadcast logo, album art, song name
Rear mount camera	• Image at rear of vehicle
CMU	• Incoming call pop-up screen display • Fuel monitor screen • Warning guidance screen display • Navigation screen display • Temperature warning pop-up screen display

- By converting the video signal to a LVDS signal, the CMU is able to display stable images in the center display virtually free from the effects of noise.
- The LVDS communication pathway between the CMU and the center display corresponds to HDCP*3.

*2 : The LVDS displays the electrical properties of the signal using low-voltage differential signaling. LVDS is a digital, wired transmission technology for the short range which adopts a differential interface having the characteristics of low voltage and low power consumption.

*3 : HDCP (High-bandwidth Digital Content Protection system) is a copyright protection technology which prevents content from being copied fraudulently. The pathway in a display device such as a display, which sends/receives digital signals from a video playback device such as a computer, is encrypted.

Audio transmission signal function

- The CMU transmits the audio signal received from a mobile device such as the following auxiliary jack/USB port/ SD card slot hub and Smartphone devices to the TAU. The CMU receives audio signals from the voice recognition microphone and performs operations according to the commands.

TA (traffic announcement) function

- If the TAU receives a TA broadcast while listening to the following modes, the CMU switches the audio to the TA broadcast automatically. In addition, the traffic alert screen is displayed in the center display.
  - FM radio
  - CD
  - USB
  - AUX
  - Bluetooth® audio
  - Aha™ radio
  - Sticher™ radio
- When the TA broadcast has finished, the audio unit returns to the mode that was being listened to before the TA broadcast. In addition, the center display cancels the traffic alert screen display and displays the screen prior to the traffic alert screen.

Traffic announcement screen

Button	Content
TA off	• Cancels the TA broadcast playback and traffic alert screen display, and displays the screen prior to the traffic alert screen. • Sets the TA (traffic announcement) function to OFF.
Cancel	Cancels the TA broadcast playback and traffic alert screen display, and displays the screen prior to the traffic alert screen.
Remove	Cancels the traffic alert screen display and displays the screen prior to the traffic alert screen.

Note

- If the TA (traffic announcement) function is set to OFF, the audio does not switch automatically to TA broadcasts even if a TA broadcast is received.

Fuel economy function

- The CMU receives the fuel consumption amount and travel distance information signal from the instrument cluster and displays it in the center display as the fuel economy monitor. For details, refer to the fuel economy monitor of the CENTER DISPLAY. (See CENTER DISPLAY [WITH CENTER DISPLAY].)

Navigation function

- The CMU performs map display and route guidance based on the position information received from the GPS antenna and the map information received from the auxiliary jack/USB port/SD card slot hub. For details, refer to the ENTERTAINMENT SYSTEM navigation function. (See ENTERTAINMENT SYSTEM [WITH CENTER DISPLAY].)

Turn-by-turn (TBT)

- The turn-by-turn (TBT) function displays the travel direction of the route guidance in the active driving display. The turn-by-turn (TBT) operates when all of the following conditions have been met. For details, refer to the navigation function turn-by-turn (TBT) of the ENTERTAINMENT SYSTEM. (See ENTERTAINMENT SYSTEM [WITH CENTER DISPLAY].)
  - — Route guidance of navigation function is operating
  - — Navigation guidance is set to ON using the personalization function for the active driving display
- The CMU sends the turn-by-turn (TBT) display request signal to the instrument cluster to display the direction heading and distance in the active driving display. (See ACTIVE DRIVING DISPLAY.)

Automatic configuration function

- When the ignition is switched ON (engine off or on) after the CMU is replaced, the CMU reads the vehicle specification information sent from the instrument cluster, and stores the vehicle specification information.

On-board diagnostic function

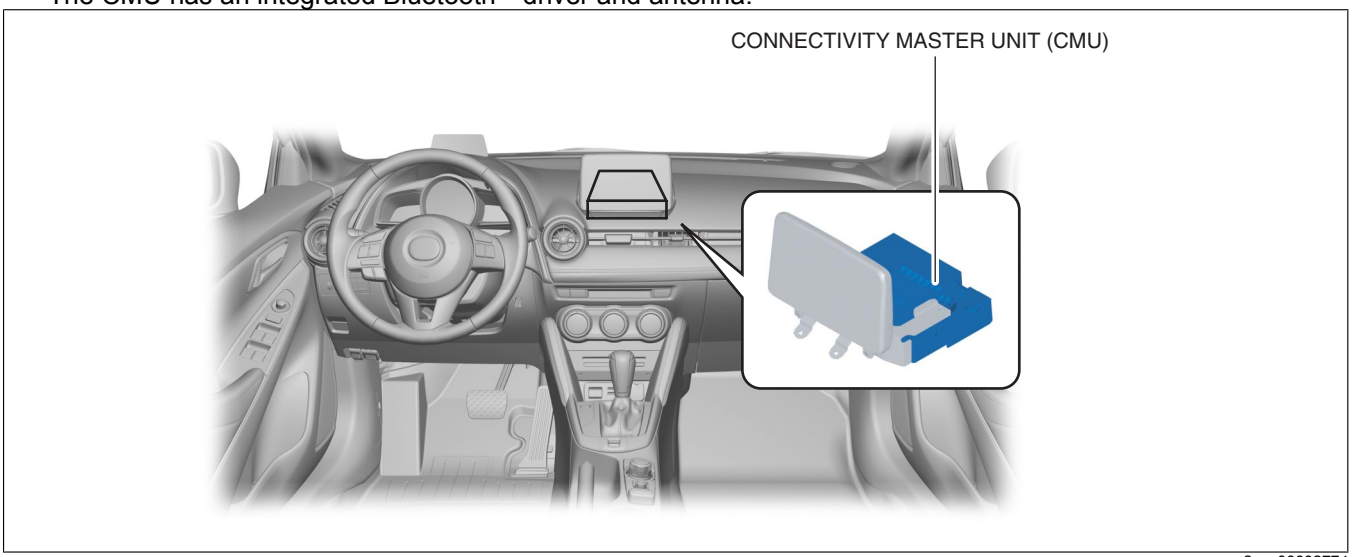
- The CMU has an on-board diagnostic function to facilitate system diagnosis. DTCs are stored in the event of a malfunction occurring. For details, refer to the DIAGNOSTIC SYSTEM. (See ON-BOARD DIAGNOSTIC [ENTERTAINMENT SYSTEM (WITH CENTER DISPLAY)].)

Diagnostic assist function

- The CMU has a diagnostic assist function which can verify the following. For details, refer to the DIAGNOSTIC SYSTEM. (See ON-BOARD DIAGNOSTIC [ENTERTAINMENT SYSTEM (WITH CENTER DISPLAY)].)
  - — Connection status of related parts
  - — Operation status of related parts
  - — Related part specifications

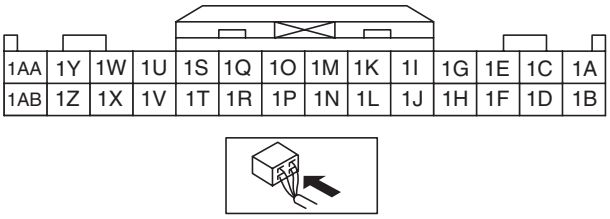
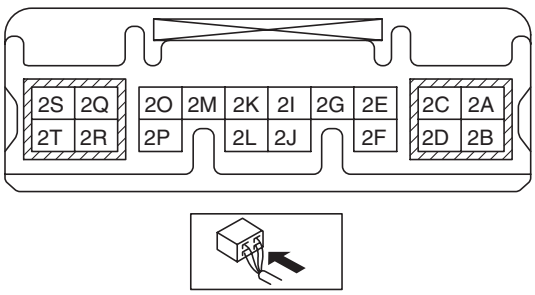
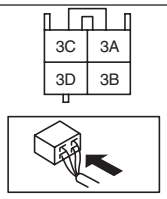
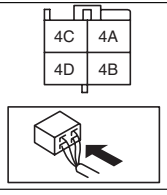
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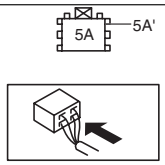
- The CMU is located centrally inside the instrument cluster.
- The CMU has an integrated Bluetooth® driver and antenna.



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Input/output signal table

Terminal		Signal
	1A	Rear mount camera power signal *4
	1B	Rear mount camera signal ground*4
	1C	Rear mount camera power supply *4
	1D	Rear mount camera power supply ground*4
	1E	—
	1F	—
	1G	DVD video signal input
	1H	DVD video signal ground
	1I	—
	1J	—
	1K	iPod video signal
	1L	iPod video signal ground
	1M	Steering1 switch input
	1N	Steering2 switch ground input
	1O	Steering3 switch input
	1P	Microphone connection detection
	1Q	Microphone power supply
	1R	Microphone power supply ground
	1S	Microphone voice input (+)
	1T	Microphone voice input (-)
	1U	—
	1V	—
	1W	—
	1X	—
	1Y	CMU audio output LH (+)
	1Z	CMU audio output LH (-)
	1AA	CMU audio output RH (+)
	1AB	CMU audio output RH (-)
	2A	—
	2B	—
	2C	Ground
	2D	Ground
	2E	HS-CAN (+)
	2F	HS-CAN (-)
	2G	—
	2I	Local HS-CAN (+)
	2J	Local HS-CAN (-)
	2K	LIN communication
	2L	Ground
	2M	—
	2O	RS485 signal (+)
	2P	RS485 signal (-)
	2Q	ACC
	2R	B+
	2S	—
	2T	—
	3A	—
	3B	LVDS (+)
	3C	—
	3D	LVDS (-)
	4A	Ground
	4B	USB data (+)
	4C	USB power supply
	4D	USB data (-)

Terminal		Signal
	5A	GPS
	5A'	Ground

*4 : With park assist system

Operation

- For the CMU operation, refer to the ENTERTAINMENT SYSTEM. (See ENTERTAINMENT SYSTEM [WITH CENTER DISPLAY].)

Fail-safe

- If the CMU receives an electrical signal (LVDS signal) of a temperature of **approx. 82** degrees C or more from the center display, it performs the following controls:
  - — Sends a temperature warning pop-up screen display request signal to the center display and displays the temperature warning pop-up screen on the LCD.
  - — Temporarily lowers the brightness of the center display LCD. This is to protect the center display LCD and the circuit.